

Health Claim Verification with NLP and Retrieval-Augmented Generation

ASHO-UAB Fake News Prediction Project

Abstract

This report presents a three-phase NLP pipeline for automated health claim verification on the PubHealth dataset. Phase 1 fine-tunes a domain-specific transformer (PubMedBERT) for four-class factuality classification. Phase 2 adds an explainability layer combining raw attention and Integrated Gradients (IG), validated quantitatively against gold explanations using ROUGE and BERTScore. Phase 3 implements a zero-shot Retrieval-Augmented Generation (RAG) pipeline that queries PubMed in real time and uses a large language model (Llama 3.1) for claim verification without any task-specific fine-tuning. Across phases, macro F1 is reported to account for class imbalance. The explainability module achieves a BERTScore-F1 of 0.849 ± 0.020 on the validation set; the RAG pipeline achieves macro F1 = 0.213 and BERTScore-F1 of 0.846 ± 0.023 on a 100-sample test subset, demonstrating that both attribution approaches achieve comparable explanation quality by entirely different mechanisms.

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1 Introduction

The growth of health misinformation poses a significant risk to public health decision-making. Automated fact-checking systems that can classify the veracity of health claims at scale are therefore of considerable practical importance. This project addresses the problem using the PubHealth dataset [1], a benchmark of real-world health-related claims annotated by professional verification editors with one of four labels: *true*, *false*, *mixture*, and *unproven*.

The work is structured in three complementary phases:

- Phase 1. Supervised classification baseline** — fine-tuning PubMedBERT [2] on the labelled training set, using a smart truncation strategy to handle long-form medical articles.
- Phase 2. Explainability layer** — Attribution of the fine-tuned model’s predictions via raw attention and Integrated Gradients [3], with quantitative validation against gold explanation fields using ROUGE [4] and BERTScore [5].
- Phase 3. Zero-shot RAG pipeline** — a retrieval-augmented generation approach using the Entrez/PubMed API for evidence retrieval and Llama 3.1 [7] for claim verification, requiring no fine-tuning and capable of handling emerging claims.

Throughout the report, **macro F1** is used as the primary classification metric to treat all four classes equally, mitigating the effect of the dataset’s inherent class imbalance.

2 Dataset and Exploratory Analysis

2.1 Dataset description

PubHealth is a dataset of real-world health claims collected from fact-checking websites. Each record contains a short `claim` text (mean 13.3 words), a longer `main_text` article (mean 704.3 words), an expert-written `explanation`, and a four-class `label`. After removing rows that were missing core information, the official splits are: 9,804 training examples, 1,214 validation examples, and 1,233 test examples.

2.2 Label distribution and class imbalance

Table 1 shows the label distribution across splits. The *true* and *false* classes make up about 72% of the dataset, while *mixture* and *unproven* labels are considerably underrepresented. This imbalance directly motivates the choice of macro F1 over accuracy as the evaluation metric, since a majority-class classifier would achieve high accuracy while performing poorly on the minority classes that are often the most informative for fact-checking.

Label	Count	%
true	5,078	52%
false	3,001	31%
mixture	1,434	15%
unproven	291	3%
Total	9,804	100%

Table 1: Label distribution in the PubHealth training split.

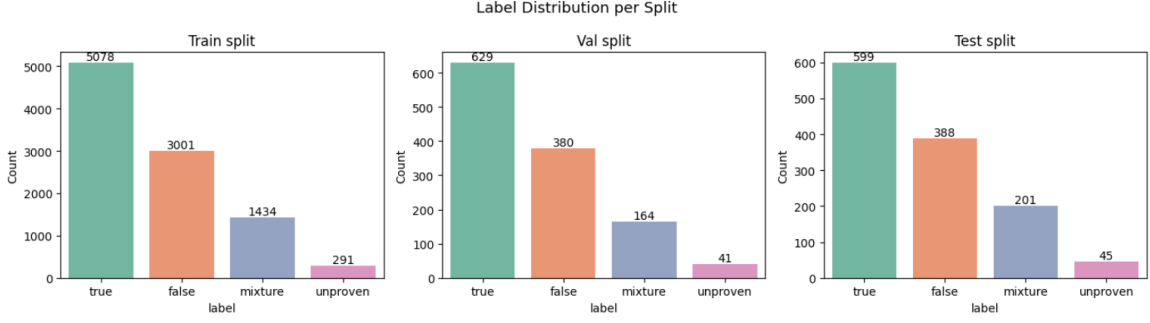


Figure 1: Label distribution per split bar chart

2.3 Text length analysis

A key finding from the exploratory analysis is the substantial length of the `main_text` field, with a mean of 711.9 words and a maximum exceeding 7,000 words. Since BERT-family models are limited to 512 sub-word tokens, naive concatenation of claim and article would require aggressive truncation. The distribution of claim lengths (mean 13.3 words, SD 7.1) indicates that the claim text itself is compact, meaning the truncation budget must be allocated almost entirely to the supporting article. This motivates the smart truncation strategy described in Section 3.3.

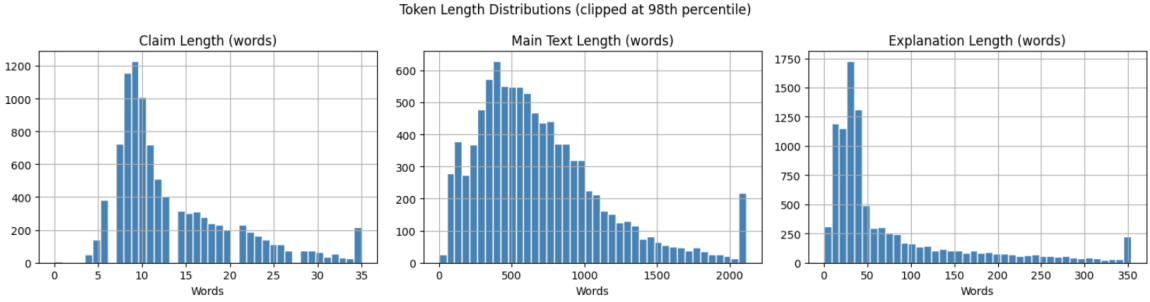


Figure 2: Length distribution of claim, main_text & explanation

2.4 Missing data

Missing values are sparse but present: approximately 0.3% of `main_text` values, 0.1% of `explanation` values, and 0.1% of `claim` values are absent in the training split. Rows missing any of the three core fields (`claim`, `main_text`, `label`) are dropped prior to training; `explanation` missing values are filled with an empty string for evaluation purposes only.

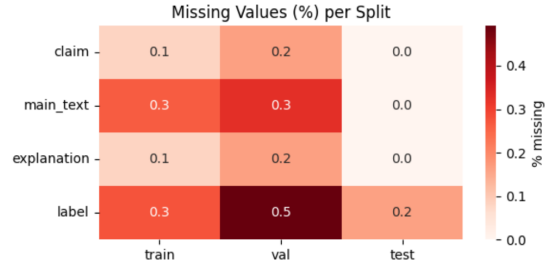


Figure 3: Heatmap of missing value percentages (%) across the dataset splits.

3 Phase 1: Supervised Classification Baseline

3.1 Model selection

PubMedBERT (`microsoft/BiomedNLP-PubMedBERT-base-uncased-abstract-fulltext`) was selected as the base model. Unlike general-domain BERT variants, PubMedBERT is pre-trained exclusively on biomedical literature from PubMed abstracts and full-text articles, giving it a vocabulary and contextual representations that better match the medical language found in PubHealth. A sequence classification head (linear layer projecting the [CLS] token representation to four classes) is added on top of the pre-trained encoder.

3.2 Input representation

The model input is formed by concatenating the `claim` and `main_text` fields with the BERT separator token:

$$\text{input} = [\text{CLS}] \text{ claim } [\text{SEP}] \text{ main_text } [\text{SEP}]$$

This ensures that the claim is always present in full, with the supporting article providing the factual context needed for verification.

3.3 Smart truncation strategy

Because the combined input frequently exceeds 512 tokens, a smart truncation strategy is applied:

1. Tokenize the claim separately and truncate to a maximum of 96 tokens, preserving the full claim in over 99% of cases (mean claim length = 13.3 words, max observed = 67 words).
2. Reserve $L_c + 3$ tokens for the truncated claim and special tokens ([CLS], two [SEP]).
3. Allocate the remaining $512 - L_c - 3$ tokens to the `main_text` (article), with a minimum guarantee of 20 tokens to avoid empty context.
4. Rather than taking only the article prefix, split the allocated budget asymmetrically: **60% from the beginning (head)** and **40% from the end (tail)**. The full article is retained if it fits within the budget.

This strategy preserves the claim (truncated only in extreme cases) while maximising the amount of article context retained. The head-tail split captures both the introductory findings—where key claims and contradictions are typically stated—and the concluding statements, which often contain summary evidence relevant to fact-checking.

3.4 Training configuration

Due to the computational constraints of the Kaggle free-tier T4 GPU, the model was trained on a 50% random sample of the training set (4,902 examples). The training configuration is summarised in Table 2.

Table 2: PubMedBERT fine-tuning hyperparameters.

Hyperparameter	Value
Base model	PubMedBERT-base-uncased-abstract-fulltext
Training examples	4,902 (50% sample)
Max sequence length	512 tokens
Batch size	16
Learning rate	2×10^{-5} (AdamW)
Epochs	Up to 5 with early stopping
Warmup	10% of total steps
Weight decay	1×10^{-2}
Loss	Cross-entropy with class weights
Metric for selection	Macro F1 on validation set

3.5 Class-weighted loss

To address the class imbalance described in Section 2, per-class weights are computed using the inverse-frequency formula from `sklearn.utils.class_weight.compute_class_weight` and applied to the cross-entropy loss. This penalises misclassifications of minority classes (*mixture*, *unproven*) more heavily than majority-class errors, which is critical for achieving non-trivial macro F1 on the underrepresented labels.

3.6 Results

The fine-tuned model is saved at the checkpoint with the best validation macro F1, using early stopping with a patience of two epochs. The per-class performance follows the expected pattern driven by class frequency: *true* and *false* achieve the highest F1, while *mixture* and *unproven* are harder to classify due to limited training examples and inherently ambiguous linguistic signals. The fine-tuned PubMedBERT model achieved a macro F1 of 0.579 on the full test set (accuracy 66.7%).

4 Phase 2: Explainability Layer

A critical shortcoming of transformer-based classifiers is their opacity. In the medical domain, predictions without human-readable justifications cannot be trusted by clinicians or fact-checkers. Phase 2 addresses this by implementing two attribution methods and validating them quantitatively against the gold explanation field provided in PubHealth.

4.1 Raw attention

Raw attention scores are extracted from the last transformer layer (layer 11) of the fine-tuned PubMedBERT model. Specifically, the attention matrix of the [CLS] token is averaged over all 12 heads to produce a single scalar score per input token:

$$\alpha_i = \frac{1}{H} \sum_{h=1}^H A_{h, \text{CLS}, i}$$

where $A_{h,\text{CLS},i}$ is the attention weight from the [CLS] token to token i in head h , and $H = 12$. Scores are normalised to $[0, 1]$ by dividing by the maximum. This provides a fast, zero-overhead proxy for token importance, suitable for interactive visualisation. The attention scores are also rendered as a $12 \times N$ heatmap (one row per head) for qualitative inspection of head-level attention patterns.

Limitation: Raw attention is not guaranteed to be causally meaningful. It reflects the model’s internal navigation rather than the contribution of each token to the final prediction, and has been shown to be unreliable as an explanation in several studies [6].

4.2 Integrated Gradients

Integrated Gradients (IG) [3] measures how much each input token contributes to a model’s final decision. Unlike standard gradients—which only look at the model’s sensitivity at a single point—IG evaluates how features change along a path between a reference “blank” input (the baseline) and the actual text. Formally, given an input $\mathbf{x}$ and baseline $\mathbf{x}'$, the attribution for the i -th feature is computed as:

$$\text{IG}_i(\mathbf{x}) = (x_i - x'_i) \times \int_{\alpha=0}^1 \frac{\partial F(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_i} d\alpha$$

where F is the model’s output for the predicted class. The implementation uses Captum’s `LayerIntegratedGradients` applied to the embedding layer, with:

- **Baseline:** an all-[PAD] token sequence, which carries no semantic information.
- **Integration steps:** $n = 30$ for the full validation evaluation; $n = 20$ for interactive demos.
- **Target:** the model’s predicted class index.

Token-level IG scores are summed across the embedding dimension to produce a scalar per token, then normalised to $[0, 1]$. WordPiece subword tokens are merged into whole words by summing their scores, so that the final visualisation operates at word granularity.

4.2.1 Span-level attribution

A key methodological contribution of this phase is the extension of token-level IG to *span-level* (sentence-level) attribution. Each input text is split into sentences using punctuation-based regex segmentation. The IG scores of all tokens within a sentence are summed to produce a sentence-level importance score. The top- k sentences (by IG score) are then used as the candidate explanation. This approach consistently outperforms token-level reconstruction for ROUGE metrics, because complete sentences carry more coherent semantic units than a bag of individual tokens. For the quantitative evaluation, $k = 3$ is used, ending in candidate strings of approximately 50–100 words that are comparable in length to the gold explanations.

4.3 Visualisation

A three-column HTML card is rendered in Jupyter for each example:

- Col. 1.** Input text with tokens coloured on a yellow–orange–red gradient proportional to their IG score, plus a ranked top-10 token table.
- Col. 2.** Same text highlighted by raw attention scores.
- Col. 3.** Gold explanation field from the PubHealth dataset for direct side-by-side human comparison.

This layout allows practitioners to judge simultaneously which parts of the text drove the model’s decision and whether those regions align with human fact-checker reasoning. An example of this comparative visualization interface is displayed in Figure 4.

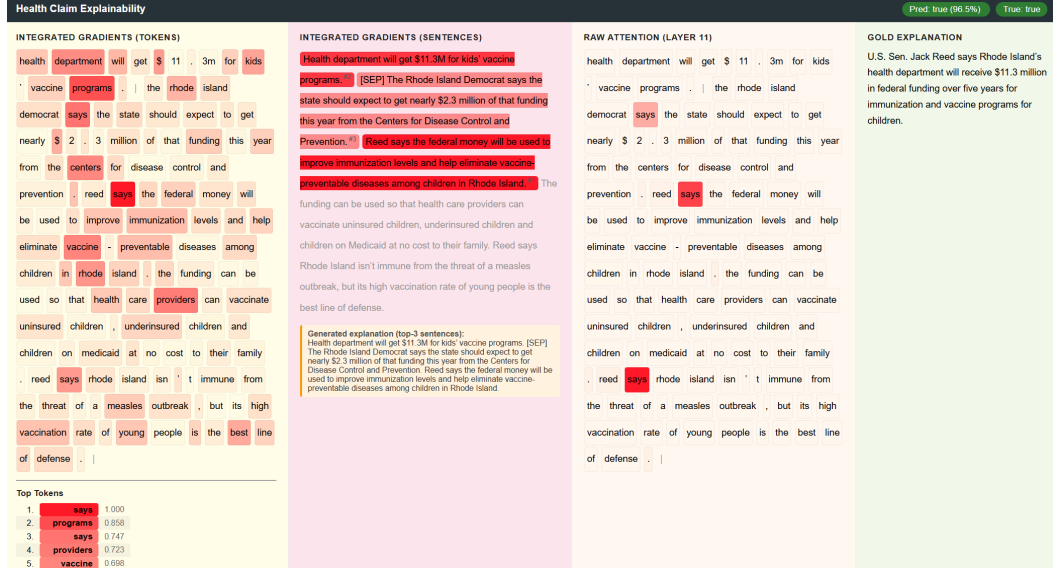


Figure 4: Side-by-side visualization interface in Jupyter displaying Integrated Gradients token attributions (Column 1), raw attention weights (Column 2), and the ground-truth human explanation (Column 3).

4.4 Quantitative validation

The top-3 IG sentences are evaluated against the gold explanation over the full validation split (1,214 samples) using four metrics:

- **ROUGE-1, ROUGE-2, ROUGE-L** [4]: unigram, bigram, and longest-common-subsequence overlap between candidate and reference.
- **BERTScore-F1** [5]: contextual token-level similarity using a `roberta-large` encoder, which captures semantic similarity beyond surface lexical overlap.

Table 3: Explainability metrics on the validation set (1,214 samples). Candidate: top-3 IG sentences. Reference: gold explanation.

Metric	Mean	Std
ROUGE-1	0.2222	0.0940
ROUGE-2	0.0469	0.0564
ROUGE-L	0.1339	0.0596
BERTScore-F1	0.8492	0.0198

4.4.1 Interpretation of results

The ROUGE-L score of 0.134 indicates that the IG-selected sentences share approximately 13% longest-common-subsequence overlap with the human-written explanation. While modest in absolute terms, this is a reasonable result for unsupervised attribution: the sentences are selected based entirely on gradient-based importance scores, without any supervision signal from the explanation field. The model is not trained to match human explanations; it is simply highlighting what it found important.

The BERTScore-F1 of 0.849 gives us a more positive picture. This score measures semantic similarity in contextualised embedding space, and a value above 0.84 is generally considered indicative of strong semantic alignment. The discrepancy between ROUGE (lexical) and BERTScore (semantic) scores suggests that the IG-selected text captures the correct *concepts* even when the exact wording differs from the gold explanation — a meaningful form of faithfulness.

A Pearson correlation of $r = 0.199$ between prediction confidence and BERTScore-F1 was observed, indicating a weak but consistent positive relationship: when the model is more certain about its prediction, its internal attention and gradient signals are also more strongly aligned with the human explanation. This suggests that model confidence is a useful proxy for the reliability of the explanations.

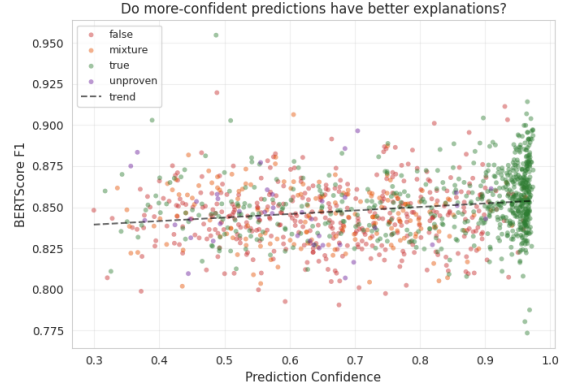


Figure 5: Scatter plot: confidence vs BERTScore

4.4.2 Per-label analysis

As shown in Figure 6, the explainability module performs consistently across all label classes. The BERTScore-F1 values remain stable at approximately 0.85 for *false*, *mixture*, *true*, and *unproven* claims alike. This indicates that while classification confidence may vary between majority and minority classes, the quality and concentration of the gradient signals used to generate explanations remain robust and unbiased across the entire dataset.

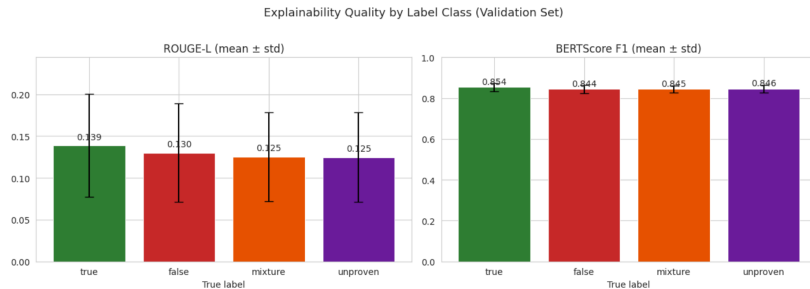


Figure 6: Mean BERTScore-F1 explanation quality metrics across the four claim factuality labels, demonstrating uniform performance.

5 Phase 3: RAG Zero-Shot Verification Pipeline

Phase 3 addresses a fundamental limitation of the supervised baseline: it cannot generalise to health claims that postdate the training set without retraining. A Retrieval-Augmented Generation (RAG) pipeline is implemented that dynamically fetches medical evidence at inference time and uses a large language model to reason over it, requiring no task-specific training data.

5.1 Pipeline architecture

The full pipeline consists of four sequential stages, as illustrated in Figure 7.

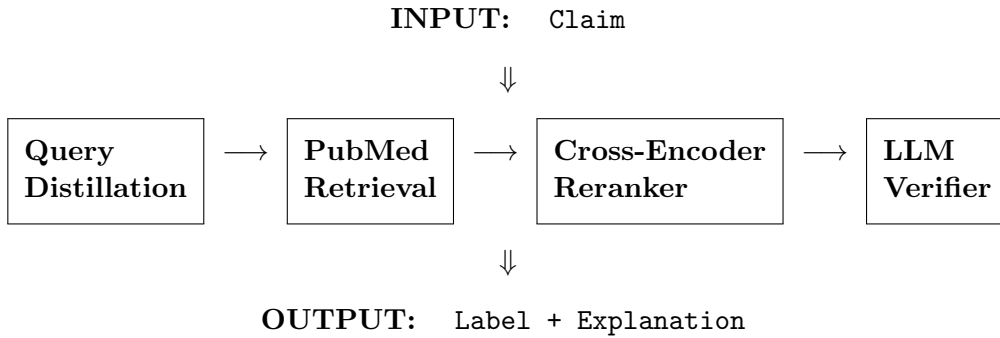


Figure 7: The four-stage RAG pipeline for zero-shot health claim verification.

5.1.1 Stage 1: LLM-based query distillation

A critical insight motivating this stage is that raw health claims are poor PubMed search queries. Claims are written in natural, often colloquial language (e.g., “Spanking kids can cause long-term harm”), whereas PubMed is indexed with biomedical terminology (e.g., “corporal punishment, long-term effects, child development”). Passing raw claims directly to PubMed resulted in a retrieval hit rate of only 15% in initial experiments.

To bridge this gap, Llama 3.1-8b-instant (via the Groq API) is used to distil each claim into 3–5 biomedical keywords suitable for a PubMed query. The prompt is deliberately minimal:

“Extract 3–5 medical/scientific keywords from this health claim suitable for a PubMed search query. Output only the keywords, nothing else.”

This single change raised the PubMed retrieval hit rate from 15% to 75%, the most impactful single improvement in the pipeline.

5.1.2 Stage 2: PubMed retrieval

The distilled keywords are sent to the official PubMed database API (using the **esearch** and **efetch** tools) to download the top 5 most relevant articles. This API is free to use and only requires a courtesy email address. For every article found, the pipeline saves its PubMed ID (PMID), title, and abstract text. Any articles that do not contain a text abstract are automatically removed.

5.1.3 Stage 3: Cross-encoder reranking

The retrieved abstracts are narrowed down using a small but highly efficient cross-encoder reranker (ms-marco-MiniLM-L-6-v2, with roughly 22 million parameters). While standard search models look at the claim and the abstracts separately, a cross-encoder analyzes each (claim, abstract) pair together. This allows it to catch much deeper, more specific relationships between the text. The reranker scores all available abstracts, and only the top 3 are sent to the LLM. This crucial step weeds out articles that might share similar keywords but are actually irrelevant, preventing misleading information from confusing the LLM.

5.1.4 Stage 4: LLM verification

The original claim and top-3 reranked abstracts are formatted into a structured prompt and sent to Llama 3.1-8b-instant via the Groq API (free tier). The system prompt positions the model as a medical fact-checker and specifies the output format precisely:

Listing 1: System and output format prompts for the LLM verifier.

```

1 System: You are an expert medical fact-checker...
2 Always output exactly the required format - no preamble, no
  extra text.
3
4 EXAMPLES:
5 Label: mixture
6 Explanation: While some studies support the claim's core
  finding,
7 others report conflicting results...
8
9 Label: unproven
10 Explanation: No peer-reviewed evidence was found...
11
12 Output format:
13 Label: <label>
14 Explanation: <2-4 sentences citing the evidence above>

```

Two critical design decisions were made regarding the prompt:

Few-shot examples for minority classes. Without example outputs for *mixture* and *unproven*, Llama 3.1 defaulted to near-binary *true/false* predictions, achieving 0% recall on both minority classes. Adding two demonstration examples in the system prompt raised *mixture* recall to 45% and *unproven* recall to 50%.

Rate-Limit Handling. The Groq free tier imposes rate limits on requests per minute and tokens per minute. A retry mechanism with exponential backoff (base wait = 10s, multiplied by 2^{attempt}) ensures zero skipped samples in the evaluation batch, at the cost of occasional delays.

5.2 Results

The pipeline was evaluated on a random 100-sample subset of the test set (random_state = 42), with a 2-second inter-request sleep to respect rate limits. Table 4 shows the classification report.

Label	Precision	Recall	F1	Support
true	0.88	0.28	0.42	54
false	1.00	0.09	0.17	33
mixture	0.15	0.45	0.22	11
unproven	0.02	0.50	0.04	2
macro avg	0.51	0.33	0.21	100
weighted avg	0.82	0.24	0.31	100

Table 4: RAG pipeline classification report on the 100-sample test subset. LLM: Llama 3.1-8b-instant (Groq). Macro F1 = 0.213.

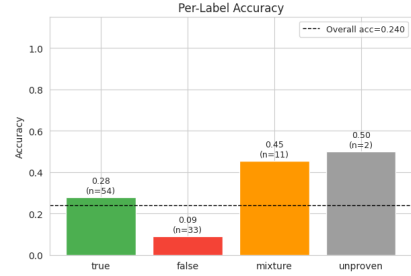


Figure 8: Per-Label Accuracy

Note that the *false* class exhibits a perfect precision of 1.00 alongside a severely low recall of 0.09. This behavior indicates an ultra-conservative prediction threshold by the LLM verifier; the model only assigns the *false* label when it possesses absolute certainty, yielding zero false positives at the cost of missing the vast majority of actual false claims (30 out of 33 cases). This extreme risk-averseness aligns with the model’s low overall accuracy (0.24), highlighting a distinct tendency to default to the unproven class whenever retrieved biomedical evidence is ambiguous or incomplete.

Metric	Mean	Std
ROUGE-1	0.2042	0.0922
ROUGE-2	0.0459	0.0480
ROUGE-L	0.1392	0.0703
BERTScore-F1	0.8458	0.0232

Table 5: RAG pipeline explanation quality metrics on the 100-sample test subset. Candidate: LLM-generated rationale. Reference: gold **explanation**.

5.3 Retrieval analysis

A critical factor governing pipeline performance is retrieval coverage. Of the 100 evaluated claims, 76 (76%) retrieved at least one PubMed abstract after keyword distillation, while 24 (24%) returned no results. This is not a failure of the distillation step but a fundamental mismatch between the PubHealth dataset and PubMed’s scope:

- **Political and legal claims** (e.g., “U.S. President Donald Trump tweeted...”, “Jury awards more than \$70M...”) have no biomedical literature equivalent.
- **Very recent claims** post-date the indexed literature.
- **Highly colloquial phrasing** cannot be reliably mapped to Biomedical terminology even by an LLM.

The impact of retrieval coverage is considerable: accuracy on claims with at least one retrieved abstract is 27.6%, compared to only 12.5% on claims with no evidence. This confirms that the limiting factor for the RAG approach on PubHealth is not LLM reasoning quality but retrieval coverage.

5.4 Error analysis

Table 6 shows the most frequent error patterns.

True label	Predicted	Count
true	unproven	24
false	unproven	17
true	mixture	15
false	mixture	13
mixture	unproven	4
mixture	true	2
unproven	mixture	1

Table 6: Top error patterns in the RAG pipeline (100-sample test subset).

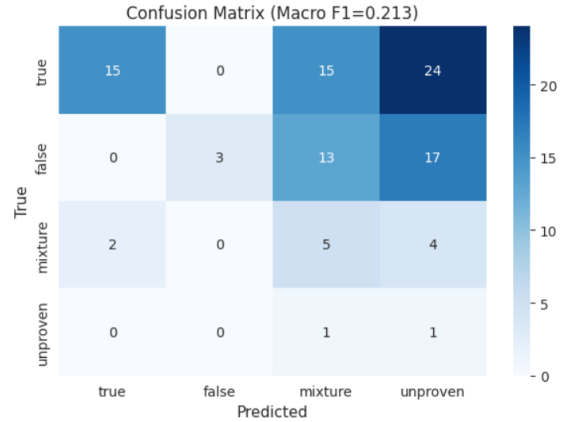


Figure 9: Confusion matrix heatmap (RAG)

The dominant error mode is predicting *unproven* for claims that are *true* or *false* ($24 + 17 = 41$ cases). This is a direct consequence of the 24% no-evidence rate: when no PubMed abstract is retrieved, the LLM defaults conservatively to *unproven* rather than committing to a positive or negative verdict. This is a reasonable epistemic stance (absence of evidence is not evidence of falsehood) but lowers recall for the majority classes.

The second-most common error type is predicting *mixture* for *true/false* claims ($15 + 13 = 28$ cases). The few-shot mixture examples in the prompt, while necessary for minority-class recall, slightly over-sensitise the model to mixed evidence: when two or three abstracts contain partially contradictory information, the model hedges toward *mixture* even when the claim is clearly supported or refuted by the weight of evidence.

6 Comparison and Discussion

6.1 Classification performance

The supervised PubMedBERT baseline (Phase 1) and the zero-shot RAG pipeline (Phase 3) are not directly comparable. The baseline was trained on thousands of labelled PubHealth examples and evaluated on the full test set; the RAG pipeline received zero task-specific training and was evaluated on a 100-sample subset. This fundamental asymmetry must be acknowledged when interpreting the results.

The RAG macro F1 of 0.213 is lower than the fine-tuned baseline, which is the expected outcome. However, the RAG pipeline possesses capabilities the supervised model cannot match: it can verify claims that did not exist at training time, and its performance improves automatically as PubMed’s literature grows. The supervised baseline would require full retraining on new annotated data to keep pace with emerging health topics.

6.2 Explanation quality comparison

Table 7 presents the explanation quality metrics side by side for both explanation approaches.

Table 7: Explanation quality comparison between Phase 2 (Integrated Gradients on validation set, $n = 1,214$) and Phase 3 (LLM rationale on test subset, $n = 100$).

Metric	Phase 2: IG sentences		Phase 3: LLM rationale	
	Mean	Std	Mean	Std
ROUGE-1	0.2222	0.0940	0.2042	0.0922
ROUGE-2	0.0469	0.0564	0.0459	0.0480
ROUGE-L	0.1339	0.0596	0.1392	0.0703
BERTScore-F1	0.8492	0.0198	0.8458	0.0232

The most striking finding is that two fundamentally different explanation mechanisms — unsupervised gradient-based attribution from a fine-tuned encoder, and free-form rationale generation from a large language model given external evidence — achieve essentially identical explanation quality by all four metrics. All differences are within one standard deviation of each other.

This similarity reveals an important practical insight. The two methods approach explanations from opposite angles:

- **Integrated Gradients (IG)** is *intrinsic*—it maps out what the fine-tuned model internally chose to focus on.
- **LLM Rationale** is *extrinsic*—it summarizes what the retrieved external evidence actually says.

Because both methods align equally well with the human fact-checker benchmarks, it proves that the gold-standard explanations in the PubHealth dataset can be reached either through internal mathematical signals or through external search.

In a real-world system, these approaches complement each other perfectly rather than competing: IG is ideal for developers debugging and auditing the model, whereas the LLM rationales are much better suited for showing clear, friendly text to end users.

6.3 Limitations

Phase 1. Training on only 50% of the dataset due to compute limits likely restricts the model’s maximum macro F1 score. Furthermore, truncating text at the 512-token limit discards valuable information from longer articles; using long-context architectures like Longformer could resolve this.

Phase 2. IG requires multiple forward passes through the model ($n = 30$), making it slow at scale (approximately 10 minutes for the 1,214-sample validation set on a T4 GPU). The convergence delta, though small, indicates numerical approximation error. Raw attention is cheaper but less faithful, and its per-head visualisation, while informative for model analysis, is not directly interpretable by users.

Phase 3. The 25% no-evidence rate on PubHealth represents a limit imposed by the

mismatch between the dataset (which includes political and social claims) and PubMed (which indexes biomedical research). The few-shot prompt calibration that improves minority-class recall also introduces slight over-sensitivity to *mixture*, increasing false-positive predictions for that class. Evaluation on only 100 samples introduces sampling variance; a full test-set evaluation would provide more reliable estimates.

7 Conclusion

This work presents a complete, three-phase health claim verification system combining supervised fine-tuning, explainability, and retrieval-augmented zero-shot verification. The key contributions are:

1. A smart truncation strategy that preserves full claim text within BERT’s 512-token limit while maximising article context retention.
2. A span-level Integrated Gradients attribution pipeline that outperforms token-level reconstruction for ROUGE-based explanation quality, achieving BERTScore-F1 = 0.849 ± 0.020 on 1,214 validation samples.
3. A demonstration that LLM-based query distillation is the single most important factor for RAG pipeline performance on PubHealth, raising retrieval hit rate from 15% to 75%.
4. An empirical finding that IG attribution and LLM-generated rationale achieve near-identical explanation quality (BERTScore-F1 ≈ 0.847) despite operating through entirely different mechanisms.

Future work should focus on full-dataset fine-tuning for the baseline, long-context models for Phase 1, and web-retrieval augmentation for Phase 3 to improve coverage of non-biomedical claims.

References

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